3.6 Choosing Model

Supervised Learning, Unsupervised Learning, Bias–Variance Tradeoff, and Loss Functions

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I. Supervised Learning

Supervised learning refers to training a model using input-output pairs. The model learns a mapping $f : X \to Y$ from labeled training data, with the goal of making accurate predictions on unseen data.

Categories:

- **Classification:** Predicts discrete classes (e.g., spam detection, medical diagnosis).
  Examples: Logistic Regression, SVM, Decision Trees, Random Forest, Naive Bayes.
- **Regression:** Predicts continuous values (e.g., house price prediction).
  Examples: Linear Regression, Ridge/Lasso, SVR.

Key Properties:

- Requires labeled data
- Trained to minimize prediction error (loss)
- Evaluated using metrics like Accuracy, F1, RMSE, etc.

II. Unsupervised Learning

Unsupervised learning uses unlabeled data. The goal is to discover hidden patterns, clusters, or lower-dimensional structures.

Categories:

- **Clustering:** Group data points based on similarity.
  Examples: K-Means, DBSCAN, Hierarchical Clustering.
- **Dimensionality Reduction:** Reduce data features while preserving structure.
  Examples: PCA, t-SNE, Autoencoders.

Key Properties:

- Works without labels
- Objective: structure discovery
- Evaluation: Silhouette score, explained variance, visual inspection

III. Overfitting, Underfitting, and Bias–Variance Trade-off

Bias–Variance Decomposition:

$$\text{Expected Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

Bias: Error from simplifying assumptions in the model (e.g., linearity). High bias leads to underfitting.

Variance: Error from sensitivity to training data. High variance leads to overfitting.

Underfitting:

- Model too simple to capture data complexity
- High bias, low variance
- Poor performance on both training and test data

Overfitting:

- Model too complex, memorizes training data
- Low bias, high variance
- Good performance on training, poor generalization

IV. Loss Function

Loss functions quantify prediction error and are used to guide training through optimization (e.g., gradient descent).

Common Loss Functions:

- **Mean Squared Error (MSE):** For regression

$$\mathcal{L}_{\text{MSE}} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

- **Mean Absolute Error (MAE):**

$$\mathcal{L}_{\text{MAE}} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

- **Binary Cross-Entropy:** For binary classification

$$\mathcal{L}_{\text{CE}} = -\sum_{i=1}^{n}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)]$$

- **Categorical Cross-Entropy:** For multi-class classification

Choosing a Loss Function:

- Use MSE or MAE for regression tasks
- Use Cross-Entropy for classification
- Match the loss function to your task and evaluation metric

V. Conclusion

Choosing the right model requires understanding the problem type (supervised vs. unsupervised), the complexity of the data, and trade-offs in generalization.

Key Takeaways:

- Supervised models need labeled data and predict known outputs.
- Unsupervised models explore structure without labels.
- Avoid underfitting (high bias) and overfitting (high variance) by selecting appropriate model complexity.
- Use loss functions that reflect your objective and help train models effectively.

Final Note:
Model selection is both a science and an art — use cross-validation, domain knowledge, and evaluation metrics together to make informed decisions.